

Executive Summary

In this project, I evaluated model performance and decision-making strategies across claim severity prediction and fraud detection workflows.

For claim severity modeling, Model B consistently outperforms Model A for personal auto claims, reducing prediction error by about \$1,085 per claim with stable results across folds. However, for commercial property claims, although Model B shows a larger average improvement, the results are more variable and not statistically reliable. Based on this, I recommend adopting Model B for personal auto, but not for commercial property at this time.

For fraud triage in personal auto, a cost-based framework indicates that claims should be investigated when the predicted probability of fraud exceeds 10%, balancing investigation cost and fraud risk.

Finally, regarding workers' compensation claims, analysis of real outcomes shows that a lower threshold of approximately 5% minimizes total cost, reducing losses by about \$23,500 per 1,000 claims. This suggests the current 10% threshold is too conservative.

Overall, these results show that validating model decisions with real data can significantly improve cost efficiency.

Request 1: Claim Severity Model Comparison

Business Question

Northbrook Mutual currently uses Model A to predict claim costs, with Model B proposed as a replacement. The goal is to determine whether Model B provides a meaningful and reliable improvement.

Approach

Model performance was evaluated using mean absolute error (MAE). Improvement was defined as the difference in MAE between Model B and Model A, and consistency was assessed across cross-validation folds.

Key Findings

For personal auto claims, Model B reduces prediction error by approximately \$1,085 per claim. Despite some variability, the improvement is consistent across most folds, and the confidence interval lies entirely below zero, indicating a statistically reliable improvement.

For commercial property claims, Model B shows a larger average improvement (~\$1,600 per claim), but results are less consistent across folds, and the confidence interval crosses zero, indicating the improvement is not statistically reliable.

Recommendation

I recommend adopting Model B for personal auto claims, with high confidence due to consistent and reliable performance. For commercial property, I recommend holding off due to uncertainty in results.

Request 2: Fraud Triage (Personal Auto)

Business Question

The goal is to determine whether investigating a claim is worth the cost, given a predicted fraud probability. Investigating a legitimate claim costs \$300, while missing a fraudulent claim costs \$2,700.

Decision Logic

Using expected cost, the optimal threshold occurs where the cost of investigating equals the cost of not investigating. This results in a threshold of 10%, meaning claims with predicted fraud probability above 10% should be investigated.

Recommendation

I recommend investigating this claim, as its predicted fraud probability of 22% exceeds the 10% threshold.

More broadly, I recommend adopting a probability-based rule: investigate claims where predicted fraud probability exceeds 10%. This approach aligns decisions with cost trade-offs and minimizes expected losses. This recommendation is supported with high confidence, as it directly follows from the cost structure.

Request 3: Workers' Comp Threshold Review

Business Problem

The current fraud investigation threshold of 10% is based on cost assumptions, but observed costs are higher than expected. The goal is to determine whether this threshold is truly cost-optimal.

Key Findings

Analysis of 2,000 claims shows that the 10% threshold is not optimal. Total cost is minimized at approximately 5%, reducing cost from \$403,500 to \$356,500. This corresponds to savings of about \$47,000 per 2,000 claims, or \$23,500 per 1,000 claims.

This indicates the current threshold is too conservative and leads to under-investigation of potentially fraudulent claims. The discrepancy between the theoretical and observed optimal threshold suggests the model may not be perfectly calibrated.

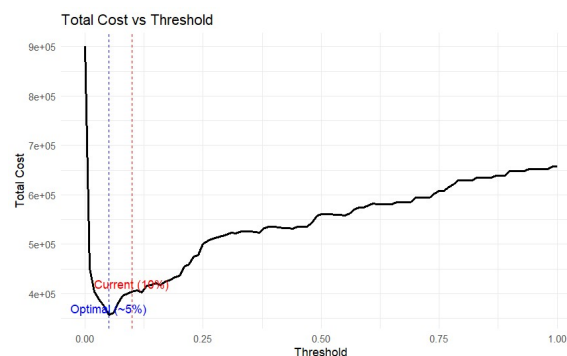


Figure 3: Total Cost vs Threshold on Workers Compensation Fraud Claim Dataset

Recommendation

I recommend lowering the threshold from 10% to 5% for workers' compensation claims. This change is expected to reduce costs by approximately \$23,500 per 1,000 claims by improving detection of high-cost fraud.

While this will increase investigation volume, the additional cost is outweighed by reduced fraud losses. I also recommend monitoring performance and recalibrating the threshold over time. This recommendation is supported with high confidence based on observed outcomes.

Appendix: Request 1 (This appendix provides supporting calculations and methodology)

Appendix A1: Evaluation Metric

Model performance was evaluated using Mean Absolute Error (MAE), which measures the average absolute difference between predicted and actual claim costs.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAE is used because it is easy to interpret in dollar terms and directly reflects prediction accuracy.

Appendix A2: Improvement Calculation

Model improvement was defined as the difference in MAE between Model B and Model A:

$$\text{Improvement} = MAE_B - MAE_A$$

A negative value indicates that Model B has lower errors and therefore performs better than Model A.

Appendix A3: Cross-Validation Approach

Model performance was evaluated using **k-fold cross-validation**, where the dataset is split into multiple subsets (folds). Each fold is used as a validation set while the remaining data is used for training.

This approach allows for:

- More robust performance evaluation
- Assessment of variability across different data splits
- Reduced risk of overfitting

Appendix A4: Confidence Interval Interpretation

To assess the reliability of model improvement, a confidence interval was constructed based on fold-level results.

- If the confidence interval lies entirely **below zero**, Model B consistently outperforms Model A
- If the confidence interval **crosses zero**, the improvement is not statistically reliable

This helps determine whether observed performance differences are meaningful or due to random variation.

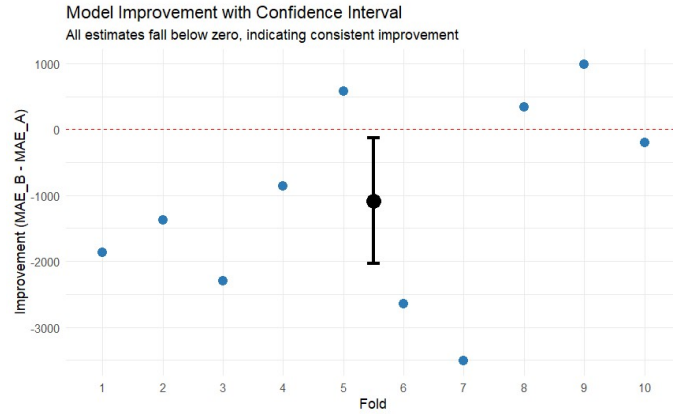


Figure 1: Confidence Interval of Model B vs Model A on Personal Auto Claims

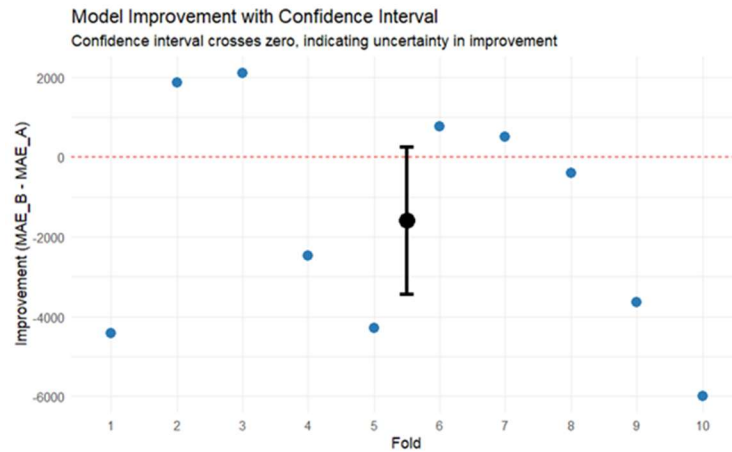


Figure 2: Confidence Interval of Model B vs Model A on Commercial Property Claims

Appendix A5: Fold-Level Results

fold <int>	MAE_A <dbl>	MAE_B <dbl>	improvement <dbl>
1	15216.60	10797.730	-4418.8669
2	15511.83	17375.823	1863.9905
3	13968.38	16073.933	2105.5506
4	15363.64	12890.915	-2472.7275
5	14841.69	10558.393	-4283.2982
6	14404.22	15158.753	754.5332
7	14810.64	15308.937	498.3014
8	14193.34	13786.767	-406.5744
9	13590.00	9951.186	-3638.8157
10	13751.88	7759.853	-5992.0282

Table 1: Fold-Level Results for Model A & Model B

Appendix A5: R Code for Model Evaluation

```
library(dplyr)

mae_by_fold <- p_auto %>%
  group_by(fold) %>%
  summarise(
    MAE_A = mean(abs(true_cost - pred_A)),
    MAE_B = mean(abs(true_cost - pred_B))
  )
mae_by_fold
```

A tibble: 10 × 3

fold <int>	MAE_A <dbl>	MAE_B <dbl>
1	3971.223	2100.8928
2	4043.414	2662.2150
3	3990.364	1691.0201
4	3836.189	2981.6898
5	3752.355	4335.6944
6	3947.866	1297.1596
7	3900.847	393.9754
8	3980.482	4320.9915
9	3772.307	4761.9119
10	3904.724	3700.6486

```
mae_by_fold <- mae_by_fold %>%
  mutate(
    improvement = MAE_B - MAE_A
  )
mae_by_fold
```

A tibble: 10 × 4

fold <int>	MAE_A <dbl>	MAE_B <dbl>	improvement <dbl>
1	3971.223	2100.8928	-1870.3303
2	4043.414	2662.2150	-1381.1990
3	3990.364	1691.0201	-2299.3438
4	3836.189	2981.6898	-854.4988
5	3752.355	4335.6944	583.3390
6	3947.866	1297.1596	-2650.7061
7	3900.847	393.9754	-3506.8712
8	3980.482	4320.9915	340.5091
9	3772.307	4761.9119	989.6046
10	3904.724	3700.6486	-204.0751

```

1 {r}
2 summary_stats <- mae_by_fold %>%
3   summarise(
4     mean_MAE_A = mean(MAE_A),
5     mean_MAE_B = mean(MAE_B),
6     mean_improvement = mean(improvement),
7     sd_improvement = sd(improvement)
8   )
9
10 summary_stats

```

A tibble: 1 × 4

mean_MAE_A <dbl>	mean_MAE_B <dbl>	mean_improvement <dbl>	sd_improvement <dbl>
3909.977	2824.62	-1085.357	1507.189

```

1 # Confidence Interval
2 {r}
3 n <- nrow(mae_by_fold)
4
5 se <- summary_stats$sd_improvement / sqrt(n)
6
7 lower <- summary_stats$mean_improvement - 2 * se
8 upper <- summary_stats$mean_improvement + 2 * se
9
10 c(lower, upper)

```

[1] -2038.5874 -132.1269

```

207 {r}
208 library(dplyr)
209
210 mae_by_fold <- c_property %>%
211   group_by(fold) %>%
212   summarise(
213     MAE_A = mean(abs(true_cost - pred_A)),
214     MAE_B = mean(abs(true_cost - pred_B))
215   )
216 mae_by_fold
217

```

A tibble: 10 × 3

fold <int>	MAE_A <dbl>	MAE_B <dbl>
1	15216.60	10797.730
2	15511.83	17375.823
3	13968.38	16073.933
4	15363.64	12890.915
5	14841.69	10558.393
6	14404.22	15158.753
7	14810.64	15308.937
8	14193.34	13786.767
9	13590.00	9951.186
10	13751.88	7759.853

```

29 # Compute model improvement by fold
30 {r}
31 mae_by_fold <- mae_by_fold %>%
32   mutate(
33     improvement = MAE_B - MAE_A
34   )
35 mae_by_fold
36

```

A tibble: 10 × 4

fold <int>	MAE_A <dbl>	MAE_B <dbl>	improvement <dbl>
1	15216.60	10797.730	-4418.8669
2	15511.83	17375.823	1863.9905
3	13968.38	16073.933	2105.5506
4	15363.64	12890.915	-2472.7275
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8	14193.34	13786.767	-406.5744
9	13590.00	9951.186	-3638.8157
10	13751.88	7759.853	-5992.0282

```

290 {r}
291 summary_stats <- mae_by_fold %>%
292   summarise(
293     mean_MAE_A = mean(MAE_A),
294     mean_MAE_B = mean(MAE_B),
295     mean_improvement = mean(improvement),
296     sd_improvement = sd(improvement)
297   )
298
299 summary_stats
300

```

A tibble: 1 × 4

mean_MAE_A <dbl>	mean_MAE_B <dbl>	mean_improvement <dbl>	sd_improvement <dbl>
14565.22	12966.23	-1598.994	2914.632

1 row

```

# Confidence interval
{r}
n <- nrow(mae_by_fold)

se <- summary_stats$sd_improvement / sqrt(n)

lower <- summary_stats$mean_improvement - 2 * se
upper <- summary_stats$mean_improvement + 2 * se

c(lower, upper)

```

[1] -3442.3687 244.3816

Appendix: Request 2 (This appendix provides supporting calculations for the fraud investigation decision rule.)

Appendix B1: Cost Structure

The decision to investigate a claim is based on two possible outcomes:

- Investigating a legitimate claim (false positive) results in a cost of **\$300**
- Failing to investigate a fraudulent claim (false negative) results in a cost of **\$2,700**

Correct decisions (investigating fraud or ignoring legitimate claims) incur no cost.

Appendix B2: Expected Cost Formulation

Let p denote the predicted probability of fraud.

- Expected cost of **investigating**:

$$(1 - p) \times 300$$

- Expected cost of **not investigating**:

$$p \times 2700$$

Appendix B3: Threshold Derivation

The optimal decision threshold occurs when the expected costs of the two options are equal:

$$(1 - p) \times 300 = p \times 2700$$

Solving for p :

$$\begin{aligned} 300 &= 3000p \\ p &= 0.10 \end{aligned}$$

Appendix B4: Decision Rule

Based on this threshold:

Investigate the claim if the predicted probability of fraud exceeds 10%.

Appendix: Request 3 (This appendix provides supporting calculations and methodology for identifying the cost-optimal fraud investigation threshold.)

Appendix C1: Cost Structure

For each claim, cost is determined based on the investigation decision and actual outcome:

- Investigate + Not Fraud → **\$500**
- Investigate + Fraud → **\$0**
- Do Not Investigate + Fraud → **\$4,500**

- Do Not Investigate + Not Fraud → \$0

Appendix C2: Decision Rule

For a given threshold t :

- Investigate if predicted fraud probability $> t$
- Do not investigate otherwise

Appendix C3: Cost Calculation

- For each claim, cost is assigned based on the decision and actual outcome. Total cost is computed as:
- Total Cost = $\sum_{i=1}^n \text{Cost}_i$
- where Cost_i represents the cost incurred for claim i .

Appendix C4: Threshold Optimization Procedure

To identify the optimal threshold:

- Evaluate thresholds from 0 to 1 in increments (e.g., 0.01)
- Apply the decision rule at each threshold
- Compute total cost across all claims
- Select the threshold that minimizes total cost

Appendix C5: Summary Results

claim_id <int>	true_label <int>	pred_prob <dbl>	decision <chr>	cost <dbl>
41	0	0.0118	no_investigate	0
42	0	0.0179	no_investigate	0
43	0	0.1018	investigate	500
44	0	0.0173	no_investigate	0
45	1	0.0654	no_investigate	4500
46	1	0.0635	no_investigate	4500
47	0	0.1124	investigate	500
48	0	0.0002	no_investigate	0
49	0	0.0030	no_investigate	0
50	0	0.0000	no_investigate	0

Table 2: Summary Results

Appendix C6: R Code for Threshold Optimization

The following code evaluates total cost across thresholds from 0 to 1 and identifies the threshold that minimizes overall cost.

```

thresholds <- seq(0, 1, by = 0.01)
results <- data.frame(threshold = thresholds, total_cost = NA)
for (i in seq_along(thresholds)) {
  t <- thresholds[i]
  df_temp <- df %>%
    mutate(
      decision = ifelse(pred_prob > t, "investigate", "no_investigate"),
      cost = case_when(
        decision == "investigate" & true_label == 0 ~ 500,
        decision == "investigate" & true_label == 1 ~ 0,
        decision == "no_investigate" & true_label == 1 ~ 4500,
        decision == "no_investigate" & true_label == 0 ~ 0
      )
    )
  results$total_cost[i] <- sum(df_temp$cost)
}
results[which.min(results$total_cost), ]

```

	threshold <dbl>	total_cost <dbl>
6	0.05	356500